

DEviRank: An Evidence-Weighted Network Proximity Framework for Drug Prioritization in Network-based Drug Repurposing

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Abstract. Drug repurposing aims to identify new therapeutic uses for existing drugs, but network-based prioritization can be affected by statistical thresholds, null-model estimation, and differences in drug-target set size. We present DEviRank, an evidence-weighted network proximity framework that combines statistically filtered drug-disease proximity with protein-protein interaction confidence and curated drug-gene interaction evidence. Drug scores are normalized by the number of contributing target genes to reduce target-set-size effects.

We evaluated the stability of proximity estimates across increasing random-sampling depths and assessed the impact of normalization on drug ranking. Normalization preserved the overall ranking structure while substantially reducing the dependence of drug scores on the number of contributing genes.

Primary sclerosing cholangitis was used as a case study. DEviRank identified candidates associated with disease-relevant immune, gut-liver, epigenetic, and fibrotic pathways. The results also showed that network rank does not necessarily correspond to clinical suitability. DEviRank is therefore intended as an interpretable framework for prioritizing drug-disease hypotheses for downstream biological evaluation.

Keywords: Network-based drug repurposing, Drug ranking, Network proximity analysis, Protein-protein interaction networks, Drug prioritization, DEviRank.

1 Introduction

Drug repurposing aims to identify new therapeutic uses for existing drugs, offering a cost-effective alternative to de novo drug development. The increasing availability of large-scale biological data has enabled computational approaches to systematically explore drug-disease associations.

Among these, network-based methods have emerged as a central paradigm, representing biological systems as protein–protein interaction (PPI) networks. In this framework, diseases are modeled as modules of interacting proteins, and drugs are evaluated based on the proximity of their targets to disease-associated proteins. Previous studies have shown that drugs whose targets are closer to disease modules are more likely to have therapeutic effects.

A widely used approach is the proximity measure introduced by Guney et al., which computes the average shortest distance from drug targets to the closest disease-associated proteins. While effective, this framework has several limitations. First, it excludes proteins without recorded interactions (isolated proteins), which can bias distance estimation. Second, it relies on fixed proximity thresholds that are not explicitly tied to statistical significance. Third, it does not incorporate interaction confidence into the final ranking.

In this work, we introduce DEviRank, an evidence-weighted network proximity framework for drug prioritization. DEviRank extends classical proximity analysis by (i) retaining isolated proteins to ensure unbiased distance computation, (ii) applying statistically grounded significance thresholds based on null models, and (iii) integrating PPI confidence and curated drug–gene interaction (DGI) evidence into a unified ranking score.

The proposed method is disease-agnostic and designed for interpretability and reproducibility. By combining network topology with interaction evidence and statistical validation, DEviRank provides a robust and transparent framework for network-based drug prioritization.

2 Related Work

Network-based drug repurposing is rooted in the network-medicine paradigm, in which diseases are viewed as perturbations of interconnected molecular systems rather than as consequences of isolated genes or proteins [1]. Within this framework, drug–disease relationships can be evaluated by determining how closely drug targets are positioned to disease-associated proteins in the human protein–protein interaction network.

A widely used implementation of this concept was introduced by Guney et al. [2], who systematically compared several interactome-based distance measures for quantifying drug–disease proximity. Among the evaluated measures, the closest-distance metric (d_c), defined as the average shortest-path distance from each drug target to its nearest disease-associated protein, provided the strongest discrimination between known and unknown drug–disease associations [2]. This framework established network proximity as an interpretable approach for relating drug targets to disease modules and has subsequently been applied in multiple drug-repurposing settings.

More recent computational approaches have extended beyond single-layer PPI proximity by integrating multiple types of biomedical relationships, including drug–target, gene–disease, pathway, phenotype, and adverse-effect information, within heterogeneous networks and biomedical knowledge graphs [3]. Representation-learning approaches, including network embeddings and graph neural networks, can exploit these

multirelational data to predict previously unobserved drug–target or drug–disease associations [3]. These methods increase representational capacity but also introduce additional model complexity and can make the biological basis of individual predictions less directly interpretable; consequently, explainability remains an important consideration in knowledge-graph-based drug repurposing [3].

DEviRank follows a complementary strategy. Rather than replacing explicit network proximity with a learned latent representation, it retains an interpretable PPI-based proximity framework and extends it through statistically defined candidate filtering and evidence-weighted ranking. The method therefore does not require a disease-specific labeled training set and preserves a direct connection between candidate prioritization, network topology, and the biological interaction evidence contributing to the final score.

3 DEviRank Method

DEviRank builds directly on the network-based drug efficacy screening framework introduced by Guney et al. [2]. Guney et al. compared multiple measures of distance between drug targets and disease-associated proteins and found that the closest-distance measure (d_c) provided the best discrimination between known and unknown drug–disease associations. In this formulation, (d_c) represents the average, across drug targets, of the shortest-path distance from each drug target to its nearest disease-associated protein [2].

DEviRank adopts this closest-distance formulation as the basis for statistical drug–disease proximity assessment and extends the framework through methodological modifications to target representation and null-model evaluation, followed by an evidence-weighted drug-ranking procedure. In particular, DEviRank combines statistically filtered network proximity with interaction-confidence information and curated drug–gene interaction evidence to prioritize candidates after proximity-based selection. A worked example illustrating the calculation of drug–disease network proximity according to the Guney et al. framework is provided in the **Supplementary File**.

3.1 Network Representation and Inputs

Biological systems are modeled as protein–protein interaction (PPI) networks, in which proteins (gene products) are represented as nodes and interactions between proteins are represented as edges. Disease- and drug-associated targets are defined at the gene level but analyzed at the protein level. The PSC-associated disease target genes used as input for the DEviRank analysis are provided in **Supplementary Table 1**.

Protein–protein interactions were obtained from the STRING database [4]. In DEviRank, the PPI network is represented as an undirected graph with confidence-weighted edges. Only high-confidence STRING interactions with a combined confidence score ≥ 0.7 were retained for the analysis.

Drug–gene interaction data were obtained from DGIdb version 5.0 [5]. DGIdb interaction scores were used to represent the available evidence associated with individual

drug–gene interactions. These scores were normalized to the interval $[0,1]$, and interactions with normalized scores < 0.01 were excluded from the analysis.

3.2 Proximity Measure and Statistical Assessment

Effect of gene set definition of disease-drug distance measurement. Network-based disease–drug proximity methods assume that disease-associated and drug target genes are well represented in the protein–protein interaction (PPI) network. In practice, some proteins lack recorded interactions, and their treatment directly affects distance-based measurements.

In the framework of Guney et al. [2], such proteins are excluded. While simplifying computation, this alters target set composition and can bias the average closest distance (d_c), potentially leading to false-positive or false-negative associations.

To avoid this, DEviRank retains all disease and drug target proteins in the PPI network, ensuring consistent treatment and preventing artificial distortion of network distances.

The effect of isolated proteins depends on their role: A protein without interactions contributes distance zero only when it is shared between the drug and disease target sets; otherwise, a drug target without interactions has no finite path to disease proteins. Ignoring these cases alters proximity estimate.

This effect is illustrated in **Fig.1** using a toy PPI network. In **Fig.1a**, the drug target set $S = \{s_1, s_2, s_3\}$ and the disease target set $T = \{t_1, t_2, t_3\}$ are fully connected through the network, and all drug targets have finite paths to disease proteins. The resulting d_c between sets S and T is 2. In **Fig.1b**, an additional drug target protein s_4 without interactions is introduced. Its distance to the disease set becomes effectively infinite, increasing the average distance ($d_c = 26$). In DEviRank, unreachable drug–disease distances are replaced with a large finite penalty, chosen to exceed any finite shortest-path distance in the network. For the illustrative toy example, 100 was used as the penalty value to keep the calculation simple and transparent, whereas the full DEviRank implementation used 20,000 as the penalty value, and the network has 20000 proteins. This preserves isolated targets in the proximity calculation while ensuring that disconnected drug targets receive a strong distance penalty. In **Fig.1c**, s_4 is also a disease-associated protein t_4 , yielding distance zero and reducing the average distance ($d_c = 1.5$). These cases show that isolated proteins can either increase or decrease proximity depending on their role. Excluding such proteins changes the composition of the input target sets and can therefore alter proximity estimates.

Z-score and p-value threshold. In DEviRank, candidate associations are retained when the observed proximity satisfies $z\text{-score} \leq -1.96$ together with an empirical $p\text{-value} \leq 0.05$ obtained from degree-preserving random sampling. In contrast, Guney et al. [2] defined drug–disease pairs as proximal using an empirically selected cutoff of $z\text{-score} \leq -0.15$, chosen to provide good coverage of known drug–disease associations while limiting false-positives. This cutoff was therefore an operational classification threshold rather than a conventional statistical-significance threshold. DEviRank instead uses explicit statistical criteria to define the candidate set.

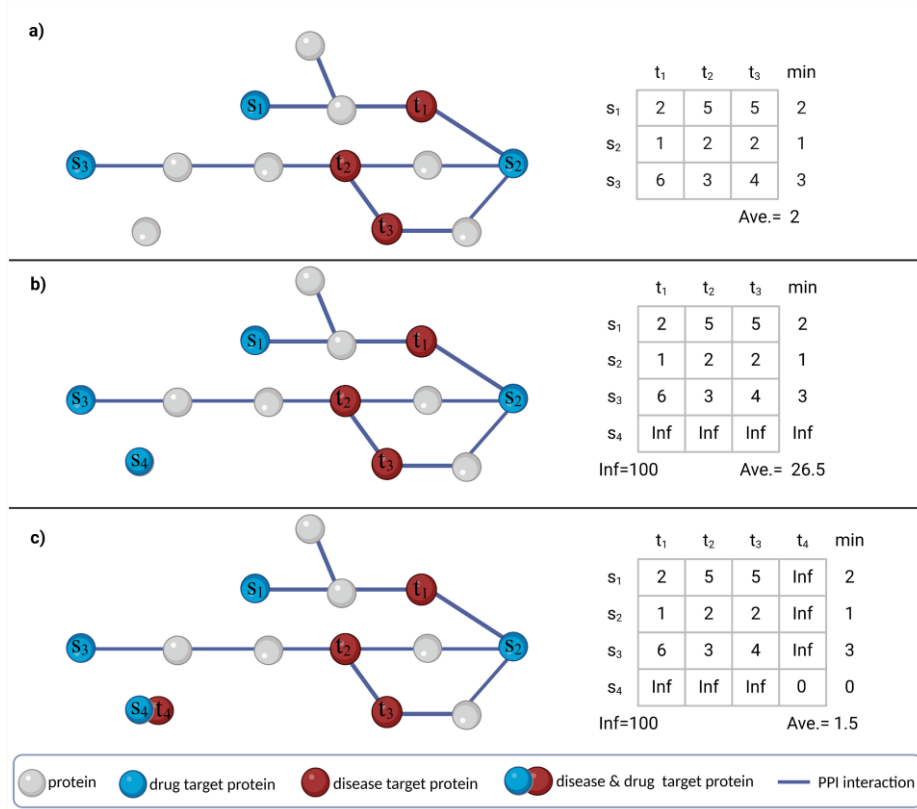


Fig. 1. It illustrates a PPI network with three possible scenarios between drug target proteins (s_i) and disease target proteins (t_i). **a)** All drug and disease proteins are connected via PPIs. **b)** shows a protein network in which one drug target protein does not have protein interactions. **c)** represents a protein network in which one drug target protein does not have protein interactions, but it is a direct target of a disease protein.

Robustness Analysis of Random Sampling Size. Following Guney et al. [2], drug relevance is assessed by comparing the observed d_c distance to a null distribution generated from degree-preserving random protein sets. A drug is considered relevant if its observed distance is significantly smaller than expected under random sampling.

The combinatorial size of the null model makes exhaustive enumeration infeasible. For example, the drug Rucaparib Camsylate targets three proteins (degrees 158, 44, and 14), with 290, 147, and 102 degree-matched candidates, yielding 4,348,260 possible target sets. Therefore, random sampling is required, but insufficient sampling can lead to unstable null estimates and unreliable z-scores.

To determine an appropriate sampling size, we evaluated 1000 (1k), 10,000 (10k), and 100,000 (100k) random samples per drug. Using $z\text{-score} \leq -1.96$ and $p\text{-value} \leq 0.05$, 39 drugs were identified with 1k sampling, 34 with 10k, and 36 with 100k. Relative to the 100k reference, the 1k configuration retained additional candidates, whereas the 10k configuration did not retain a small number of candidates identified at 100k.

To assess convergence, we compared z-scores, p-values, null means, and standard deviations using Standard Deviation of Differences (SDD), mean difference, and range, with 100k as reference. The 1k configuration shows high variability across all metrics, indicating unstable proximity estimates. In contrast, 10k closely approximates 100k, with minimal discrepancies (e.g., z-score SDD = 0.011); see **Table 1**.

This reduction in variability from 1k to 10k, and the marginal change from 10k to 100k, is consistent with $1/\sqrt{n}$ convergence predicted by the Central Limit Theorem [6]. These results indicate that 10k samples provide stable estimates, while DEviRank uses 100k samples in final analyses to prioritize statistical precision.

Table 1. Convergence analysis of random sampling sizes (1k, 10k, and 100k samples per drug). Pairwise differences in z-scores, p-values, null means, and null standard deviations are summarized using the Standard Deviation of Differences (SDD), Mean Difference, and Range of Differences. The 100k configuration serves as a high-resolution reference; smaller values indicate stronger convergence toward this reference.

	z-score	p-value	mean	SD
1k random sampling vs. 100k random sampling				
SDD	0.110	0.011	8.064	51.307
Mean Difference	0.011	0.002	0.322	0.783
Range of Differences	3.738	0.100	233.585	1000.152
10k random sampling vs. 100k random sampling				
SDD	0.011	0.003	3.038	18.165
Mean Difference	0.001	0.000	0.058	1.120
Range of Differences	0.121	0.026	77.085	349.614
1k random sampling vs. 10k random sampling				
SDD	0.110	0.010	8.982	59.364
Mean Difference	0.010	0.002	0.380	0.337
Range of Differences	3.738	0.099	249.617	1202.934

3.3 Evidence-Weighted Drug Ranking (DEviRank)

Problem Setup. Let $G = (V, E)$ denote a PPI network, where nodes represent proteins and edges represent interactions weighted by confidence scores. A disease is characterized by a set of disease target proteins $T \subseteq V$, and a drug is associated with a set of target proteins $S \subseteq V$. For each drug-protein pair (d, s) , a Drug-Gene Interaction (DGI) score quantifies the confidence and strength of the interaction.

The objective of DEviRank is to rank drugs according to their potential therapeutic relevance by integrating network connectivity between drug and disease target proteins with biological interaction evidence.

Step 1: Weighting Disease Target Proteins. Each disease-associated protein $t \in T$ is assigned a weight $w_t(t)$ reflecting its local influence within the PPI network:

$$w_t(t) = \sum_{n \in N[t]} PPI_n(1)$$

where $N[t]$ denotes the set of edges connecting protein t to its neighboring proteins.

Step 2: Weighting Drug Target Proteins. For each drug target protein $s \in S$, a network-based weight is computed by aggregating all paths connecting s to disease target proteins $t \in T$:

$$w_s(s) = \sum_{\substack{t \in T, \\ p \in P[s \rightarrow t]}} \left(\prod_{e \in p} PPI_e * (1 + w_t(t)) \right) \quad (2)$$

where $P[s \rightarrow t]$ denotes the set of paths from s to t , e represents an edge along path p , and PPI_e is the confidence score of interaction e .

Step 3: Computing the Drug Score. The final score of a drug d is obtained by aggregating the weighted contributions of its target proteins. After calculating the raw DEviRank score for each drug, scores were normalized by the number of genes contributing to the corresponding drug–disease network score. This normalization was applied because larger gene sets are expected to generate more network connections and may therefore artificially inflate raw scores:

$$w_d(d) = \left(\sum_{s \in S} DGI_t * w_s(s) \right) / n_d \quad (3)$$

Where DGI_t drug gene interaction score between drug d and gene $t \in T$. Drugs are ranked in descending order of $w_d(d)$, and n_d denotes the number of genes contributing to the drug–disease network score for drug d . Drugs were subsequently ranked according to the normalized DEviRank score.

To clarify the scoring procedure, a toy example demonstrating how the DEviRank score is calculated for an individual drug is provided in the **Supplementary File**.

Computational Complexity. Guney et al. [2] observed that effective drugs typically interact with only a subset of disease-associated proteins and that most therapeutic effects are mediated through disease proteins located at most two interaction steps away in the protein–protein interaction (PPI) network. This finding suggests that disease–drug proximity is primarily governed by local network neighborhoods rather than long-range interactions. Following this observation, DEviRank evaluates network paths between drug target proteins and disease-associated proteins by restricting path enumeration to simple paths of bounded length. In our implementation, we consider all simple paths of maximum length three, capturing both direct and indirect interactions while avoiding the combinatorial explosion associated with unrestricted path enumeration.

Let $|V|$ and $|E|$ denote the number of nodes and edges in the PPI network, $|S|$ the number of drug target proteins, and $|T|$ the number of disease target proteins. The

computation of disease protein weights scales linearly with the local neighborhood size of disease proteins, requiring $O(\sum_{t \in T} \deg(t))$ time.

The dominant computational cost arises from enumerating simple network paths of bounded length between drug target proteins and disease-associated proteins. By restricting paths to a maximum length of three, the number of candidate paths originating from a drug target protein s is bounded by $O(\deg(s)\Delta^2)$, where Δ denotes the maximum degree of the PPI network. Consequently, the overall per-drug complexity is bounded by

$$O(\sum_{t \in T} \deg(t) + \sum_{s \in S} \deg(s)\Delta^2) \quad (4)$$

and by $O(\sum_{t \in T} \deg(t) + |S|\Delta^3)$ in the worst case.

Since drug evaluations are independent, DEViRank is embarrassingly parallel across drugs, allowing efficient scaling in high-performance computing environments. The DEViRank implementation and ranked drug results and a step-by-step derivation of the time complexity and pseudocode are publicly available at **Supplementary file**.

4 Results

The experimental evaluation addressed three complementary questions: (i) whether the degree-preserving random-sampling procedure provides stable drug–disease proximity estimates; (ii) how the statistical selection criteria affect the size of the candidate set relative to the proximity framework of Guney et al. [2]; and (iii) whether normalization of the evidence-weighted DEViRank score by the number of contributing drug–target genes materially changes the ranking and reduces dependence of the score on target-set size. Primary sclerosing cholangitis (PSC) was used as a case study. Statistical filtering and score normalization were completed before literature-based biological and translational interpretation of the ranked candidates.

4.1 Robustness of proximity estimation and candidate selection

Because network-proximity statistics depend on the estimated null distribution, DEViRank was evaluated using 1,000, 10,000, and 100,000 degree-preserving random samples per drug. Using the final screening criteria of $z\text{-score} \leq -1.96$ and empirical $p\text{-value} \leq 0.05$, 39 drugs were retained with 1,000 samples, 34 with 10,000 samples, and 36 with 100,000 samples (**Supplementary Tables 2–4**). The 1,000-sample configuration showed greater variability in $z\text{-scores}$ and empirical $p\text{-values}$, whereas 10,000 samples closely approximated the 100,000-sample reference. In the convergence analysis, the standard deviation of the $z\text{-score}$ differences between the 10,000- and 100,000-sample configurations was 0.011, compared with 0.110 for the 1,000- versus 100,000-sample comparison (**Table 1**). These results indicate that 10,000 samples were sufficient to obtain relatively stable estimates in this dataset; the final analysis used 100,000 samples to obtain the highest-resolution null estimates evaluated here.

The final proximity screen retained drug–disease associations satisfying both $z\text{-score} \leq -1.96$ and empirical $p\text{-value} \leq 0.05$. Under the comparable 1,000-sample baseline

analysis, the Guney et al. proximity framework retained 951 of 2,871 screened drugs when the original operational cutoff $z\text{-score} \leq -0.15$ was applied (**Supplementary Table 5**). Guney et al. selected this cutoff to balance coverage of known drug–disease associations against false-positive rate rather than as a conventional hypothesis-testing threshold [2]. In the final DEviRank analysis, 36 drugs passed the stricter statistical screen (**Supplementary Table 6**). This comparison therefore demonstrates a difference in selectivity and decision rules, not superior predictive accuracy, because no external PSC gold-standard set was available for direct performance benchmarking.

Importantly, score normalization does not determine which drugs pass the proximity screen. The z -score and empirical p -value define the statistically retained set, whereas the evidence-weighted DEviRank score orders the retained drugs. Normalization is applied at this ranking stage.

4.2 Effect of gene-count normalization on DEviRank ranking

The raw DEviRank score is additive across the drug-target genes that contribute to the drug–disease network score. Consequently, drugs with larger contributing target sets have more opportunities to accumulate weighted network contributions. To reduce this structural dependence on target-set size, the raw score was divided by n_d , the number of contributing drug-target genes, as defined in **Eq. 3**. The normalized score therefore represents the average evidence-weighted network contribution per contributing drug-target gene rather than the total accumulated contribution.

Normalization preserved substantial overall agreement with the raw ranking but produced meaningful local reordering. Across the 36 statistically significant candidates, the raw and normalized ranks were strongly correlated (Spearman $\rho = 0.806$, $p < 0.001$). Nevertheless, the mean absolute rank change was 4.89 positions, 16 of 36 candidates (44.4%) moved by at least five ranks, and 7 of the raw top 10 remained in the normalized top 10. Thus, normalization refined rather than replaced the underlying DEviRank ordering.

The largest changes were consistent with correction for differences in the number of contributing drug genes. Vancomycin moved from raw rank 16 to normalized rank 2, olsalazine from 14 to 4, and tenecteplase from 11 to 3. Conversely, rucaparib moved from rank 8 to 20, azacitidine from 15 to 27, palbociclib from 7 to 15, panobinostat from 2 to 7, and olaparib from 4 to 9. These movements should not be interpreted as evidence that the upward-moving drugs are clinically superior; they reflect the change from an aggregate score to a per-contributing-gene score.

As a direct diagnostic of the intended normalization effect, the raw score was positively associated with the number of contributing drug-target genes (Spearman $\rho = 0.633$, $p < 0.001$), whereas this association was no longer evident after normalization (Spearman $\rho = 0.109$, $p = 0.527$). This result supports the use of the normalized score for the final DEviRank ranking because it substantially reduces the dependence of score magnitude on the number of contributing drug genes. It does not, however, imply that normalization removes all network-degree, annotation, or target-dependence biases (**Table 2**).

Table 2. Quantitative effect of drug-gene normalization on the 36 statistically significant PSC candidates.

Metric	Result
Spearman correlation: raw rank vs normalized rank	$\rho = 0.806$; $p < 0.001$
Mean absolute rank change	4.89 positions
Candidates moving by ≥ 5 ranks	16/36 (44.4%)
Top-10 overlap	7/10 (70%)
Raw score vs number of contributing drug genes	$\rho = 0.633$; $p < 0.001$
Normalized score vs number of contributing drug genes	$\rho = 0.109$; $p = 0.527$

4.3 Biological plausibility and translational interpretation of the normalized PSC ranking

The normalized DEviRank ranking was interpreted as a measure of evidence-weighted network relevance, not as a direct estimate of clinical efficacy. This distinction is particularly important in PSC, for which no approved pharmacological therapy has been shown to halt disease progression [7–9]. Candidate interpretation was therefore performed independently of the numerical score by considering PSC-relevant mechanisms, disease-specific evidence, safety, and translational feasibility. The rank numbers discussed below refer exclusively to the normalized ranking.

The highest-ranked candidate, tazemetostat, was interpreted as a strong pathway-level signal rather than as a viable PSC therapy. EZH2 activity has been linked experimentally to hepatic stellate-cell activation and fibrogenesis, and pharmacological EZH2 inhibition attenuated TGF- β -dependent stellate-cell activation and liver fibrosis in experimental models [10, 11]. However, the U.S. approval of Tazverik (tazemetostat) was withdrawn in 2026 following an increased incidence of hematologic second primary malignancies in the confirmatory trial [12]. The high normalized rank therefore supports the potential relevance of an EZH2/epigenetic axis to the PSC network, but not clinical repurposing of tazemetostat itself.

Vancomycin moved from raw rank 16 to normalized rank 2 and provides the clearest example of a high normalized rank coinciding with substantial PSC-specific investigation. PSC-IBD is a distinct intestinal inflammatory phenotype, and microbiome and gut–liver-axis mechanisms are implicated in PSC pathobiology [13, 14]. Oral vancomycin has been evaluated in PSC in randomized studies [15]. However, a phase 3, 18-month, randomized placebo-controlled trial published in 2026 found similar rates of serum alkaline-phosphatase normalization with vancomycin and placebo and no significant improvement in Mayo PSC risk score or liver stiffness; the investigators concluded that oral vancomycin was not associated with clinically significant reductions in markers of PSC disease severity [16]. Thus, vancomycin's high normalized rank is concordant with disease-specific biological and clinical interest, but it should not be presented as evidence of established disease-modifying efficacy.

Tenecteplase (rank 3) illustrates the opposite situation: a high normalized network rank with limited translational fit for chronic PSC. Experimental studies implicate the plasminogen-activation/PAI-1 axis in cholestatic liver injury and fibrosis after bile-duct ligation [17, 18], providing a possible pathway-level explanation for the network signal. Tenecteplase itself, however, is an acute thrombolytic agent [19], and no PSC-directed therapeutic study was identified in the literature reviewed for this analysis. It was therefore retained as an exploratory fibrinolysis/remodeling signal rather than a high-priority PSC therapy.

Olsalazine moved from raw rank 14 to normalized rank 4. Olsalazine is a 5-aminosalicylic-acid-based therapy with randomized clinical evidence in ulcerative colitis [20]. Its relevance to PSC is therefore most plausibly linked to the PSC-IBD phenotype and gut–liver crosstalk rather than to demonstrated reversal of biliary fibrosis. This interpretation is supported by the strong genetic relationship between PSC and IBD [21] and by evidence implicating intestinal inflammation, microbial dysbiosis, and altered bile-acid signaling in PSC-IBD pathogenesis [22]. Olsalazine was consequently retained as a phenotype-specific hypothesis rather than as a general antifibrotic PSC therapy.

Belinostat (rank 5), vorinostat (rank 6), panobinostat (rank 7), and romidepsin (rank 10), together with the panobinostat-lactate (rank 18) entry, formed a coherent HDAC-related signal. Experimental work shows that HDAC inhibition can suppress hepatic stellate-cell activation and fibrosis in bile-duct-ligation models, that HDAC8 inhibition can alleviate experimental cholestatic liver injury and fibrosis, and that combined HDAC/PDE5 inhibition reduces inflammation and fibrosis in the Mdr2-knockout model of biliary injury [23–25]. These studies support an epigenetic and fibrosis-related mechanism, but they do not establish PSC-specific efficacy for the individual oncology drugs. Accordingly, the HDAC candidates were interpreted primarily as class-level mechanistic probes. Panobinostat lactate was not treated as an independent biological mechanism from panobinostat.

Olaparib (rank 9) and rucaparib (rank 20) were retained as PARP-related stress-response signals rather than direct PSC candidates. Their prioritization may reflect network connections among DNA-damage responses, cellular stress, and inflammatory signaling, but no PSC-specific therapeutic evidence was identified. Rare immune-mediated liver injury has also been reported with olaparib [26, 27], further limiting a direct chronic-PSC repurposing interpretation.

Several immune-directed biologics - including teplizumab (rank 13), muromonab-CD3 (rank 14), blinatumomab (rank 21), catumaxomab (rank 23), mosunetuzumab (rank 28), and teclistamab (rank 29) - were not prioritized for chronic PSC treatment. Their mechanisms demonstrate that immune pathways can lie close to the PSC disease module, but the agents were developed for settings requiring potent immune intervention rather than long-term treatment of cholestatic liver disease. T-cell-engaging cancer immunotherapies can produce cytokine-release syndrome and neurotoxicity [28, 29]. In the absence of PSC-specific efficacy evidence, these compounds were therefore considered mechanistic immune-network signals rather than translational candidates.

Flucloxacillin (floxacillin, rank 12), clavulanic acid (rank 19), carbimazole (rank 17), stavudine (rank 26), and dapsone (rank 25) were deprioritized mainly because their

safety liabilities are poorly aligned with chronic PSC treatment. Amoxicillin–clavulanate is a leading cause of idiosyncratic drug-induced liver injury [30], and flucloxacillin-induced liver injury has a strong association with HLA-B*57:01 [31, 32]. Antithyroid-drug-associated hepatotoxicity has been documented for methimazole/carbimazole [33], while stavudine can cause severe mitochondrial toxicity with lactic acidosis and hepatic steatosis [34]. Dapsone has anti-inflammatory activity but clinically important hematologic, methemoglobinemic, and hypersensitivity toxicities [35]. These compounds were therefore not considered suitable PSC-focused therapies despite their network signals.

Minocycline (rank 34) and quercetin (rank 36) illustrate an important limitation of interpreting the normalized score as a complete measure of disease evidence. Minocycline remained low in the normalized ranking (rank 34) despite direct PSC investigation: a one-year open-label pilot study in 16 patients reported reductions in serum alkaline phosphatase and the Mayo PSC risk score, although the authors emphasized that the findings were too preliminary to support clinical use [36]. Minocycline is also a recognized cause of drug-induced liver injury with autoimmune features [37]. Quercetin ranked 36th but has preclinical evidence in liver-fibrosis models involving reduced macrophage recruitment and inflammatory polarization, reduced hepatic stellate-cell activation, and attenuation of fibrotic signaling [38, 39].

Fingolimod hydrochloride (rank 32) and fingolimod lauryl sulfate (rank 33) were interpreted together as an exploratory immune-trafficking signal. Fingolimod modulates sphingosine-1-phosphate receptor signaling and restricts lymphocyte egress from lymphoid tissues [40]. This creates an indirect mechanistic connection to PSC, where hepatic recruitment of CCR9-positive gut-homing lymphocytes through endothelial CCL25 has been demonstrated [41]. Fingolimod does not directly target the CCR9/CCL25 pathway, no PSC-specific therapeutic study was identified, and hepatic safety concerns are recognized [42]. The two formulation entries were therefore treated as one low-priority mechanistic signal rather than as independent supporting predictions.

The remaining statistically significant compounds - including tazarotene (rank 8), chlorhexidine hydrochloride (rank 11), palbociclib (rank 15), letrozole (rank 16), tannic acid (rank 22), penicillin V (rank 24), penicillin G potassium (rank 30), azacitidine (rank 27), hydroquinone (rank 31), and polyethylene glycol 4000 (rank 35) - were not prioritized because the literature review did not identify a sufficiently direct PSC disease-modifying rationale to outweigh limitations related to indication, mechanism, formulation, route of administration, or translational feasibility. Their presence in the statistically significant set should therefore be interpreted as a network observation requiring further hypothesis-specific validation.

Overall, the evaluation supports the normalized DEviRank score for its intended methodological purpose: reducing target-count dependence in an additive evidence-weighted network score while preserving substantial concordance with the underlying raw ranking. The literature-based PSC assessment provides examples of both concordance and discordance with the normalized ordering. High-ranked candidates can represent disease-relevant pathways without being clinically appropriate therapies, and low-ranked candidates can have prior disease-specific or preclinical evidence that is not

encoded in the DEviRank score. The normalized ranking should therefore be interpreted as a reproducible prioritization of evidence-weighted network relevance, followed by an independent biological and translational triage step, rather than as a direct prediction of therapeutic efficacy.

5 Conclusion

DEviRank is an evidence-weighted network-based drug prioritization framework that combines statistically filtered drug–disease proximity with interaction evidence and normalized drug scoring. Normalization reduces the influence of the number of contributing drug-target genes, allowing candidate ranking to reflect average evidence-weighted network relevance rather than accumulated score alone.

The PSC case study showed that DEviRank can identify biologically meaningful drug and pathway signals while also illustrating that computational rank does not directly correspond to clinical suitability. Some highly ranked compounds were mechanistically relevant but translationally unsuitable, whereas lower-ranked candidates had independent disease-specific or preclinical support.

DEviRank should therefore be interpreted as a hypothesis-generation and prioritization framework rather than a predictor of therapeutic efficacy. Its value lies in producing a statistically controlled, interpretable ranking that can guide subsequent biological and translational evaluation.

Code and data availability. All relevant codes are available on GitHub at <https://github.com/seirana/DEviRank>. The results under different sampling scenarios also available there. Supplementary file and tables are available at: <https://github.com/seirana/DEviRank/supplementary>.

Author contributions. S.H. conceived and implemented the DEviRank method, performed the analyses, developed the software, and drafted the manuscript. D.E. provided supervision and conceptual feedback. All authors approved the final version of the manuscript.

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